

Lab 3 - Written Justification

Self-Service Coffee Kiosk | PES1UG24CS245

Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice

The system is organized into Presentation, Business, and Data/Hardware layers.

Presentation handles touch input; Business handles orders and payment;

Data/Hardware handles menu pricing and receipt printing.

Reason 1 - Clear separation of responsibilities

Customer interaction, order/payment processing, and storage/hardware access are kept in separate layers, making the system easier to understand and maintain.

Reason 2 - Suitable for a focused kiosk application

The kiosk has a small, fixed workflow and one payment method. Layering avoids the operational complexity of independently deployed services.

Security Advantage

Payment processing is isolated in the Payment Service, while layer access can be restricted so components expose only the operations they require.

Performance Benefit

Direct in-kiosk component communication avoids extra network latency and distributed-service overhead. Menu/pricing data can be accessed locally.

Conclusion

The architecture provides clear boundaries, straightforward deployment, and a simple communication path for the kiosk scenario.